Bihar Engineering University, Patna
End Semester Examination - 2023

Course: B.Tech. | *Semester- II* | *Time: 03 Hours*

Code: 100204 | *Subject: Programming for Problem Solving* | *Full Marks: 70*

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Answer any seven question of the following: [2 x 7 = 14]

(a) Which of the following is a keyword used for a storage class?
(i) Printf (ii) external (iii) auto (iv) scanf

(b) Pointers are of :
(i) integer data type (ii) character data type
(iii) unsigned integer data types (iv) none of these

(c) What is the output of below program?

```
Void main (){
Chair *p= "Hello world";
Int *q;
P++;
q = (int *)p;
q--;
printf( "/n %s\n%s", p,q);}
```

(i) ello world
Ello world
(ii) Compiler error
(iii) lo world
lo world
(iv) ello world
world

(d) The minimum number of comparisons required to find the minimum and the maximum of 100 numbers is
(i) 100 (ii) 200 (iii) 150 (iv) 148

(e) Which of the following is a correct format for declaration of function?
(i) return-type function-name (argument type);
(ii) return-type function-name (argument type) {}
(iii) return-type (argument type) function name;
(iv) All of the above

(f) What is the output of this C code?

```
#include<stdio.h>
Void main ( )
{
    int b = 6;
    int c = 7;
    int a = + + b + c - - ;
}
```

(i) Run-time error (ii) 15 (iii) 13 (iv) 14

(g) Identify the correct sequence of steps to run a program.
(i) Link, load, code, compile and execute (ii) Code, compile, link, execute and load

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(iii) Code, compile, link, load and execute (iv) Compile, code, link, load and execute

(h) Why do we write (int*)before malloc?
Int*ip=(int*)malloc(sizeof(int));
(i) It is for the syntax correctness (ii) It is for the type casting
(iii) It is to inform malloc( ) function about data-type expected
(iv) None of the above

(i) What is the disadvantage of arrays in C?
(i) The amount of memory to be allocated should be known beforehand.
(ii) Elements of an array can be accessed in constant time.
(iii) Elements are stored in contiguous memory blocks.
(iv) Multiple other data structures can be implemented using arrays.

(j) What will be the output of the following code snippet?

```
#include <stdio.h>
void solve ( ){
printf ("%d ", 9 / 2);
printf ("%f" , 9.0 / 2);
}
int main ( ) {
solve( ) ;
return 0;
}
```

(i) 44.5000 (ii) 44.000 (iii) 4.5000 4.5000 (iv) 44

Q.2 (a) Discuss different looping constructs available in C. [4]
(b) Which of these constructs are exit controlled? Explain with example flowchart. [3]
(c) Elaborate the role of *break* and *continue* statements in loops. [7]

Q.3 Write a C program for any two of the following: [7x2=14]
(a) Reverse a string without using *strrev( )*
(b) GCD of two numbers
(c) Solution for *Tower of Henoi* problem.
(d) Multiplication of two matrices

Q.4 (a) How can an array of numbers be passed to function as parameter? Write an example of C program in support. [6]
(b) What do you understand by operator precedence? Explain. [4]
(c) What are the libraries? How are they different from header files? [4]

Q.5 (a) Write a program in C to find out the greatest number out of three numbers using conditional operator. [7]
(b) Explain different file opening modes. Write a program to read the content of a given file and display the number of lines and words in that file. [7]

Q.6 (a) Write a program in C to enter the records of 100 employees and display the names of those who earn more than Rs. 20,000. Create a suitable structure to keep employee records about their code, name, designation and salary. [8]

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(b) Write a program in C to print the following pattern: [6]

1
2 3
4 5 6
7 8 9 10
11 12 13 14 .....

Q.7 (a) What is dynamic memory allocation. Explain calloc( ), malloc( ), realloc( ) and free() functions with suitable examples. [6]

(b) Write a program in C using functions to swap two numbers using concept of global variables and call-by-reference concept. [8]

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Q.8 (a) Explain command line arguments in C with the help of an example. [6]

(b) Write a function in C to implement selection sort technique. [5]

(c) Which of the sorting algorithm is better than other. Explain. [3]

Q.9 Write short notes on any two of the following: [7x2=14]

(a) Enumerated data types

(b) Storage classes in C

(c) Nested if-else Vs Switch

(d) Recursion

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